

Student Name: Cuong Dang

Course: ITAI-2372-Artificial Intel Applications

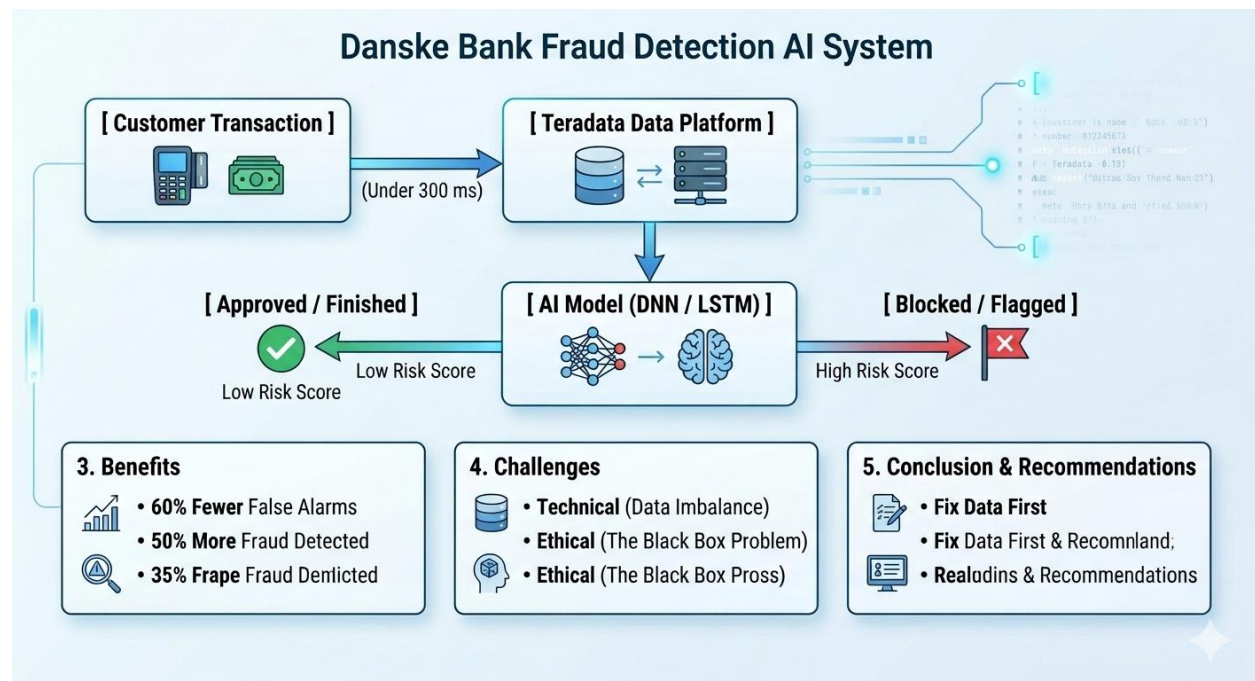
Date: June 12, 2026

A04 AI in Fraud Detection Case Study

1. Introduction

Today, digital banking is growing very fast. Criminals use smart methods to steal money online. Traditional security tools use simple rules, like blocking transactions over a specific dollar amount. These old tools are too slow and make many mistakes.

Artificial Intelligence (AI) is very important for modern banks. AI can check millions of transactions in real time. It looks at many hidden details at the same time to stop criminals immediately and protect customer trust.



2. Technology Overview

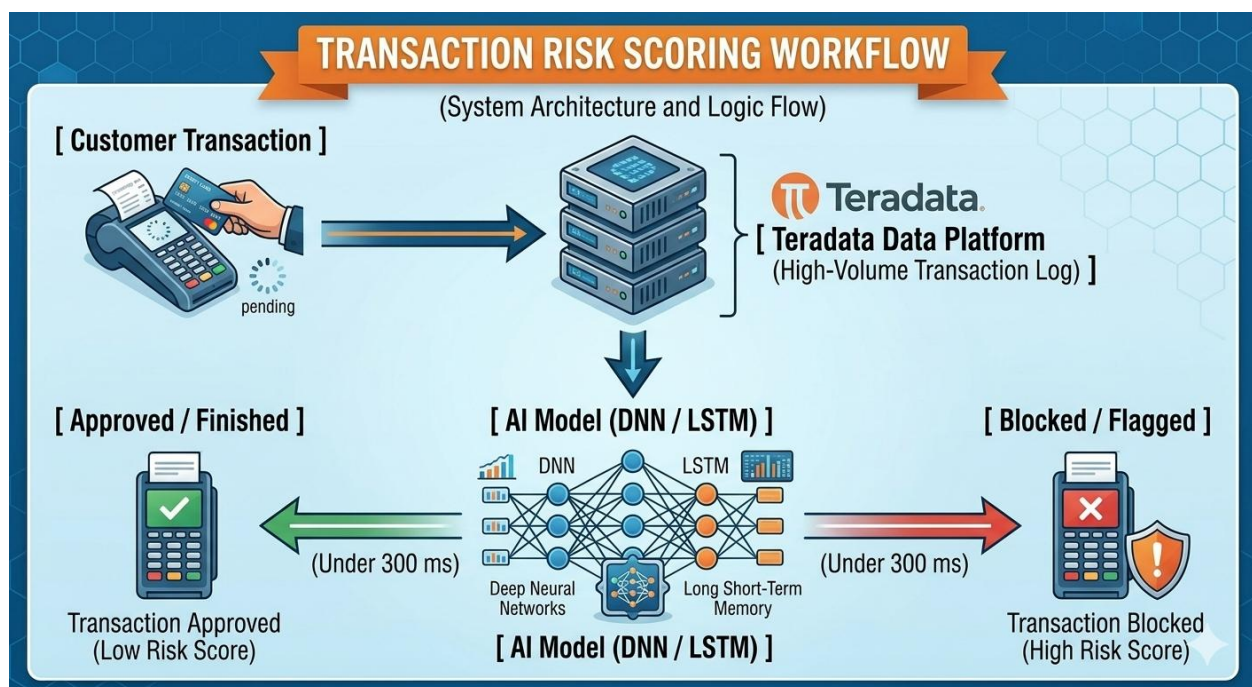
This case study looks at **Danske Bank**. They changed their old security system. They worked with a company named **Teradata** to build a new AI system using **Deep Learning** (Teradata Corporation, 2022).

Student Name: Cuong Dang

Course: ITAI-2372-Artificial Intel Applications

Date: June 12, 2026

- **Deep Neural Networks (DNNs):** This technology automatically checks transaction data like time, location, device type, and shop category without needing human help.
- **Long Short-Term Memory (LSTM):** This is a special smart model. It learns the normal bank spending habits of a customer over time. If a new transaction looks different from history, the model flags it as dangerous.



3. Benefits

Danske Bank achieved great success by using this deep learning AI system (AI.Business, 2021):

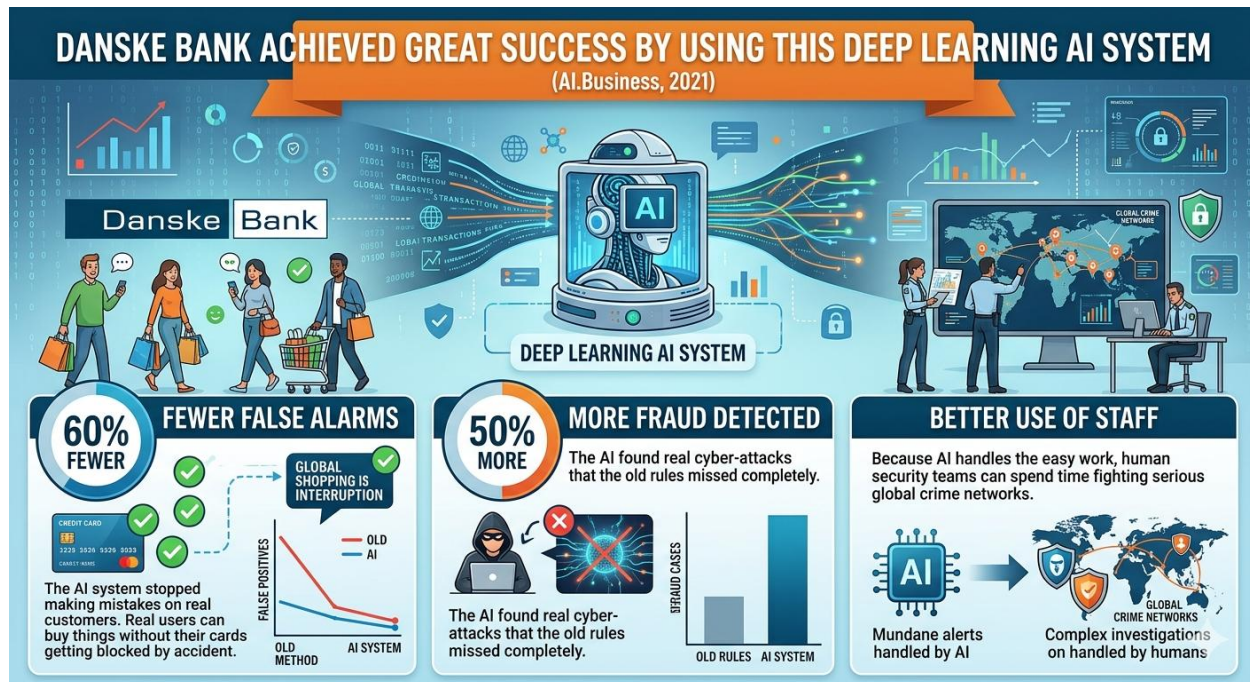
- **60% Fewer False Alarms:** The AI system stopped making mistakes on real customers. Real users can buy things without their cards getting blocked by accident.
- **50% More Fraud Detected:** The AI found real cyber-attacks that the old rules missed completely.

Student Name: Cuong Dang

Course: ITAI-2372-Artificial Intel Applications

Date: June 12, 2026

- **Better Use of Staff:** Because AI handles easy work, human security teams can spend time fighting serious global crime networks.



4. Challenges

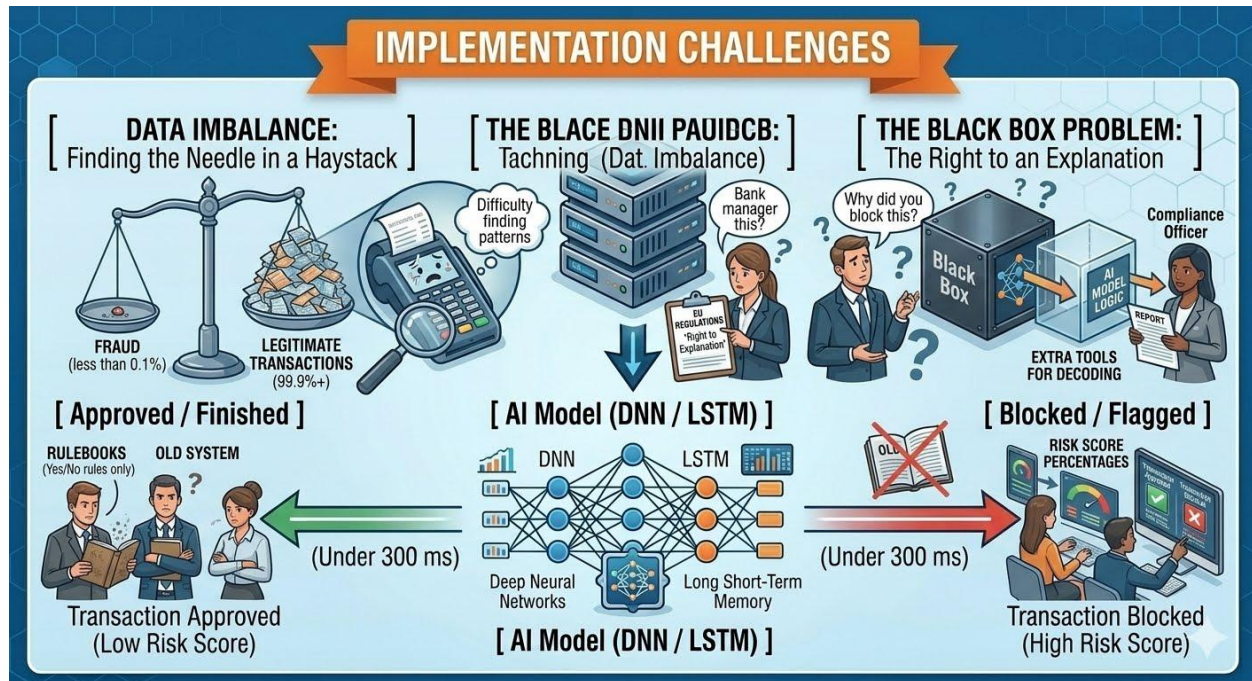
The bank faced three major problems during implementation:

- **Technical (Data Imbalance):** Real fraud is very rare (less than 0.1% of all bank traffic). It is difficult to train an AI when there are very few real examples of fraud to learn from.
- **Ethical (The Black Box Problem):** Deep learning models are very complex. They cannot easily explain *why* they blocked a user. European regulations require banks to explain their decisions, so the bank had to build extra tools to decode the AI logic.
- **Operational (Changing Mindsets):** Employees were used to simple "yes or no" rules. They had to be retrained to trust the percentage risk scores given by the AI.

Student Name: Cuong Dang

Course: ITAI-2372-Artificial Intel Applications

Date: June 12, 2026



5. Conclusion & Recommendations

Danske Bank proved that AI is much better at catching fraud than old systems. However, a bank must have clean, fast data systems before installing advanced AI.

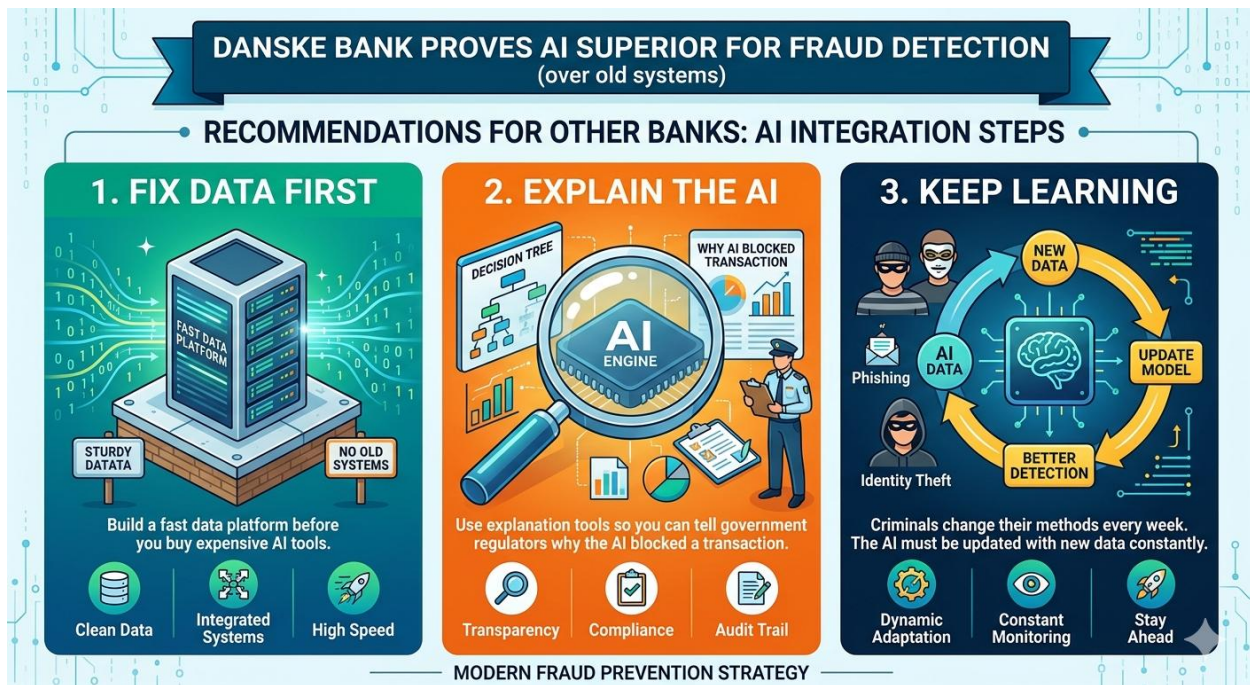
Recommendations for other banks:

1. **Fix Data First:** Build a fast data platform before you buy expensive AI tools.
2. **Explain the AI:** Use explanation tools so you can tell government regulators why the AI blocked a transaction.
3. **Keep Learning:** Criminals change their methods every week. The AI must be updated with new data constantly.

Student Name: Cuong Dang

Course: ITAI-2372-Artificial Intel Applications

Date: June 12, 2026



6. References

AI.Business. (2021). *Case study: How Danske Bank uses deep learning to defeat fraud*. AI.Business Reports.

Teradata Corporation. (2022). *Danske Bank fights fraud with real-time deep learning analytics*. Teradata Customer Success Series.